PPE CHECKPOINT

Website Blueprint

Pages, components, data, design system and build order
for the live gate monitoring dashboard

GATE 01 PROTOTYPE | FINAL YEAR PROJECT | 02 NOV 2026

WHY THIS WEBSITE EXISTS

1 Overview

The website is the monitoring screen for the PPE Checkpoint gate. It shows the current gate decision in real time, keeps a searchable history of every check, and gives a safety officer a clear compliance report. It only **reads** the data written by `main.py` (`logs/events.csv` and the `snapshots/` folder). It never changes the detection logic.

Who uses it

User	What they need
Safety officer (main user)	See at a glance if the gate is open or closed and why; review violations with photos; export a compliance report.
Admin / project team	Adjust settings, check that the system is online, run the demo.
Examiner / visitor	Watch a live demo, try the gate, and see the report update immediately.

Design principles

- **Status first.** One glance answers: open or closed, and why.
- **Words plus colour.** Every state has a text label, so it works for colour-blind viewers and from the back of a room.
- **Calm realtime.** New data animates in. Nothing jumps or re-renders the whole page.
- **Evidence for every violation.** A time, what was missing, and a photo.
- **Always demonstrable.** A built-in demo mode shows realistic sample data when no backend is running.

EVERY PAGE THE WEBSITE NEEDS

2 Site Map and Priorities

Priority tells you what to build first. **MUST** = needed for the demo. **SHOULD** = makes it feel like a real product. **LATER** = future scope; mention it in the report but do not build it now.

Page	Route	Purpose	Priority
Live Gate	/	Current decision, camera frame, KPIs, hourly chart, live log	MUST
Violation Report	/reports	Filters, summary, snapshot grid, CSV export	MUST
Event History	/events	Full searchable table of every check with filters and pagination	SHOULD
Login	/login	Sign in before viewing data (single officer account is enough)	SHOULD
Settings	/settings	Confidence, frames needed, gate open time, snapshot retention	SHOULD
About / How it works	/about	Block diagram and the decision rule, handy for the viva	SHOULD
Gates	/gates	Overview of several gates, online/offline status	LATER
404 / Offline	any	Friendly error and "backend not reachable" state	SHOULD

Navigation: a single top bar on every page: Live | Reports | Events | Gates | Settings. The active page has an orange underline. A connection badge (LIVE / OFFLINE / DEMO DATA), a clock and the user label sit on the right.

LOOK AND FEEL

3 Design System

Direction: a gate-control instrument panel. Matte basalt surfaces, hairline borders, one safety-orange accent, and signal green and red used only for status. It should feel like a product made for this gate, not a template.

Colour tokens

Token	Hex	Use
Canvas	#121316	Page background
Surface	#1A1C20	Cards and modules, with a 1px hairline border rgba(255,255,255,0.09)
Track	#2A2D33	Chart tracks, inactive fills
Accent orange	#FF6B00	Brand mark, active nav, labels, primary button
Granted	#22C55E	Access granted, compliant, OK chips
Denied (fill)	#EF4444	Borders, bars, dots
Denied (text)	#F87171	Red text on dark. Passes contrast; plain #EF4444 is borderline
Hazard yellow	#FFC400	Hazard stripe and new-row highlight only
Text	#F4F4F5	Headings and values
Muted	#A1A1AA	Labels and telemetry. About 7:1 on the surface colour

Typography

Role	Font	Size
Status word, page titles	Barlow Condensed 700, uppercase	78 px / 46 px
Numbers, times, labels	IBM Plex Mono 400/500, tabular numerals	KPI 46 px, labels 11 px with 0.14em spacing
Body and buttons	Barlow 400/500	13 to 15 px

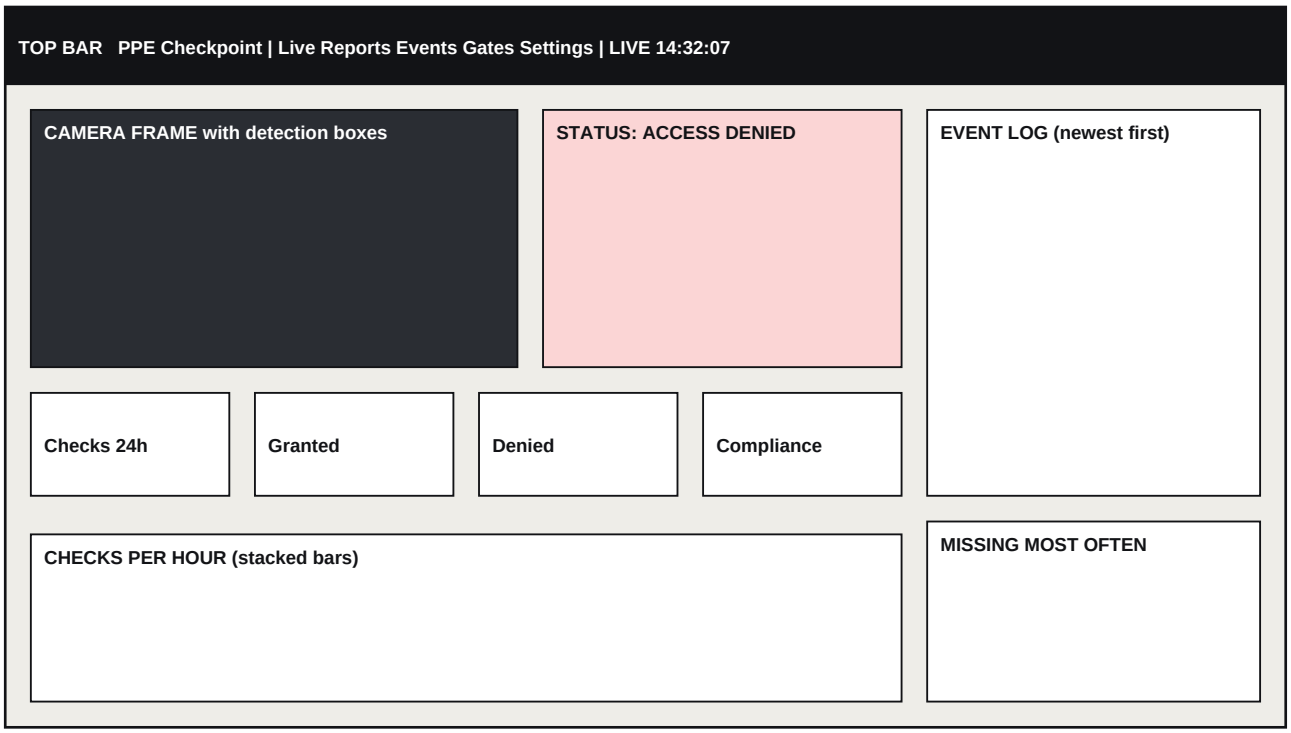
Shape and spacing: 3 to 4 px corner radius, flat colour, 1px hairlines, spacing on an 4 px grid (8, 12, 16, 20, 24, 32). Touch targets at least 44 px.

Avoid (reads as AI-made): purple or blue gradients, gradient text, frosted glass, glowing neon, emoji icons, pill-shaped cards on everything, generic hero blocks, "Welcome back" copy.

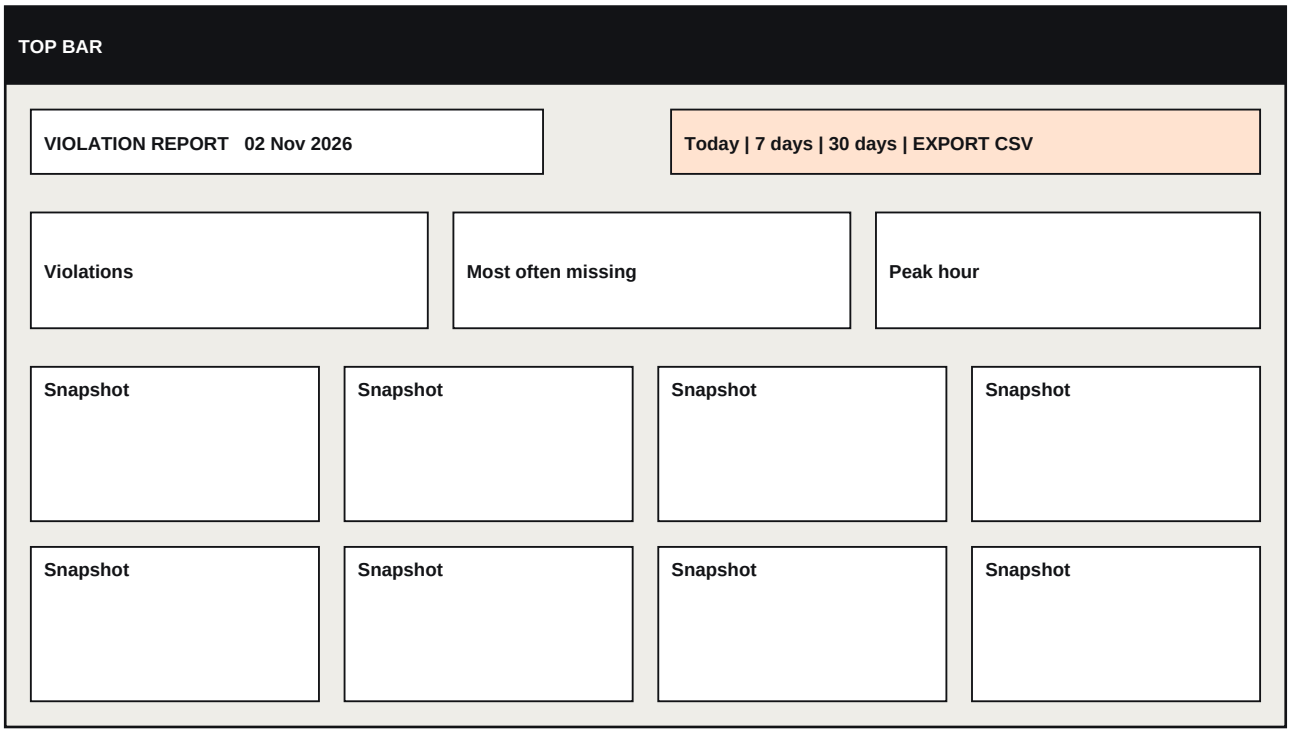
LAYOUTS AT DESKTOP WIDTH (1440 PX)

4 Page Wireframes

4.1 Live Gate (/)



4.2 Violation Report (/reports)



4.3 Event History (/events)

TOP BAR

Search | Result: All/Granted/Denied | Date range | Export CSV

TABLE Time | Verdict | Missing | Snapshot | Detail (click a row to open the photo)

Pagination: Prev 1 2 3 Next

4.4 Settings (/settings)

TOP BAR

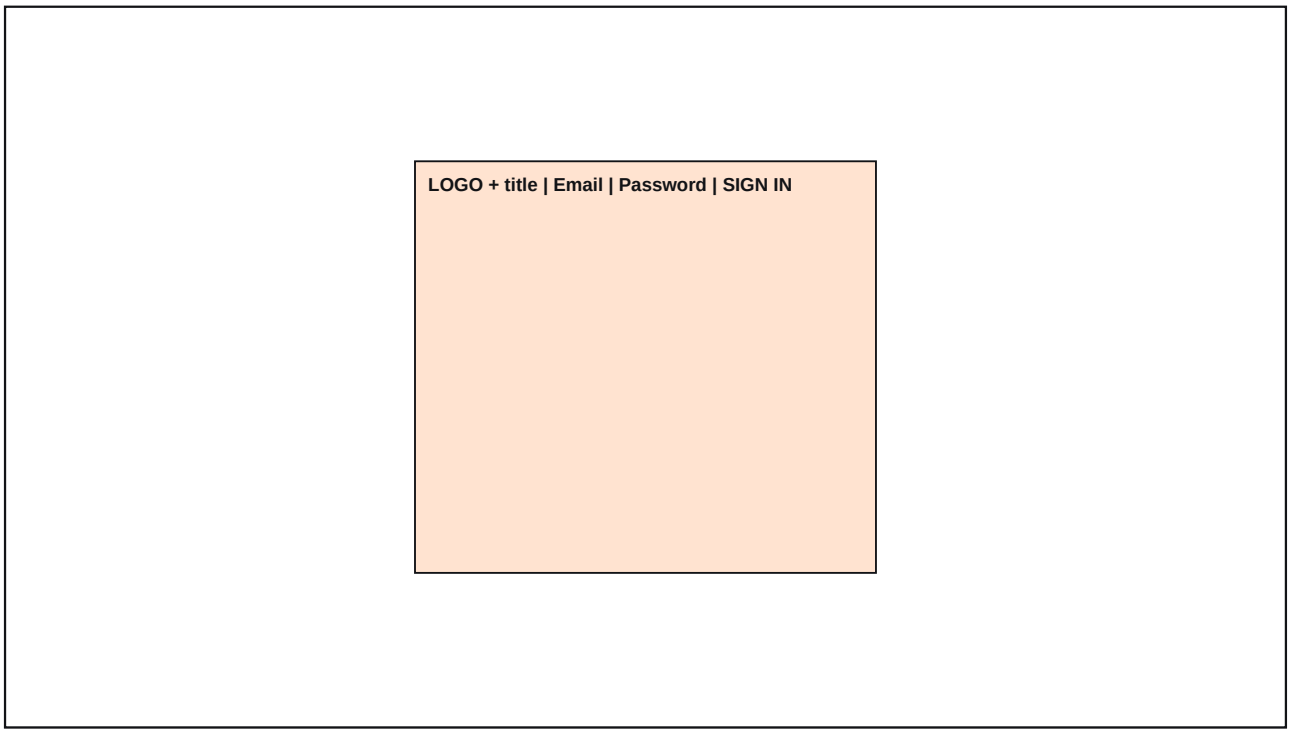
SECTIONS: Detection / Gate / Dat

DETECTION: confidence, frames needed

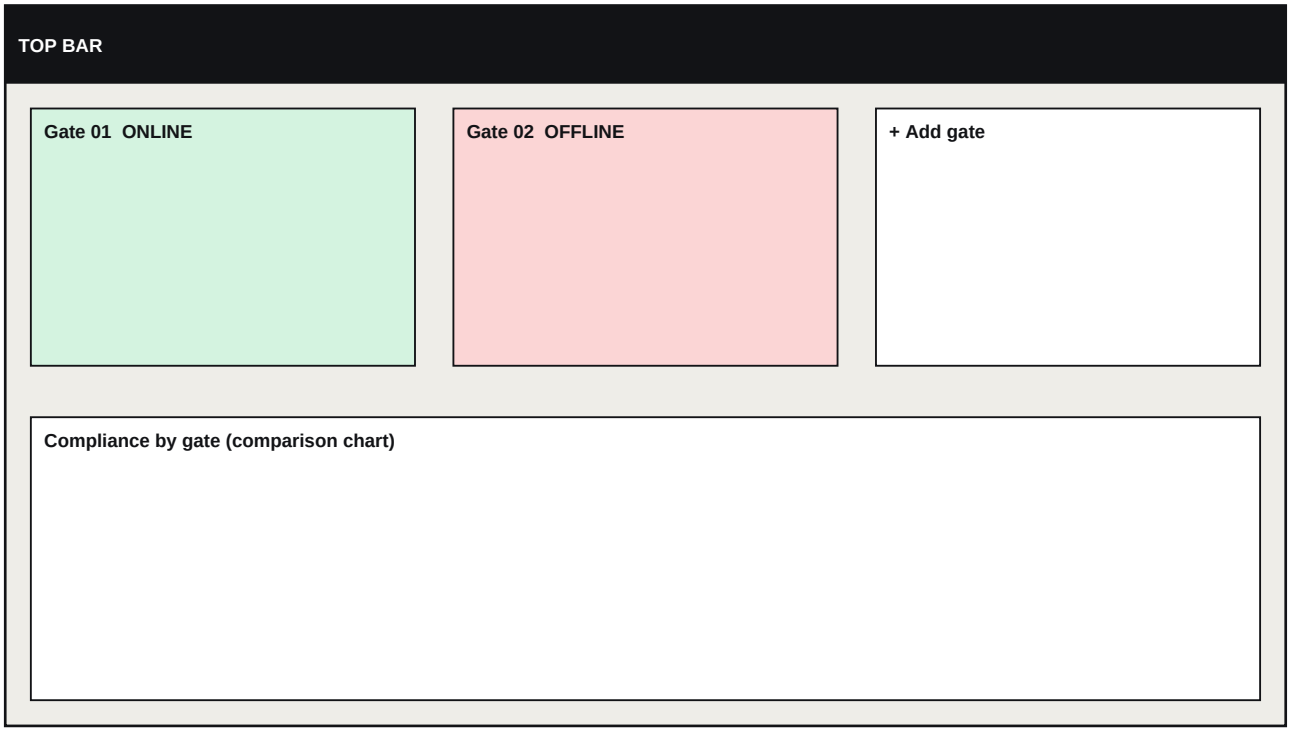
GATE: open time, deny display time

DATA: snapshot retention, clear log

4.5 Login (/login)



4.6 Gates (later) (/gates)



Wireframes are low-fidelity layout guides. The high-fidelity versions are the two mockup screens (Live Gate and Violation Report) from the design canvas.

5 Page Specifications

5.1 Live Gate (MUST)

Purpose: Show the current decision and the last few minutes of activity.

Contents

- **Status hero:** camera frame with boxes (left); status word, missing item, HELMET / VEST chips, barrier state (right). Granted = green tint, denied = red tint.
- **KPI strip:** Checks 24h, Granted, Denied, Compliance %.
- **Checks per hour:** 12 stacked bars, green granted and red denied.
- **Event log:** newest first, max 60 rows. New rows drop in with a yellow flash.
- **Missing most often:** two horizontal bars (helmet, vest).

States: Standby (nobody there), Checking, Granted (5 s), Denied (7 s), Offline, Demo data.

5.2 Violation Report (MUST)

Purpose: Give the safety officer evidence and a summary they can export.

Contents

- Title, date label, range filter (Today / 7 days / 30 days) and Export CSV button.
- Three summary tiles: violations, most often missing item, peak hour.
- Snapshot grid, 4 columns, each card shows photo, time and a chip such as NO VEST.
- Click a card to open a larger photo (lightbox) with time, items missing and file name.

States: No violations (success message), no photo (hatched placeholder), loading.

5.3 Event History (SHOULD)

Purpose: Answer "what happened at 11:40?" for any check.

Contents

- Filter bar: search text, result filter, date range, export.
- Table with sortable Time column, 25 rows per page, pagination.
- Row click opens the same lightbox as the report.

States: Empty result (no matches), loading.

5.4 Settings (SHOULD)

Purpose: Let the team tune the system without editing code.

Contents

- Detection: confidence (0.3 to 0.8), frames needed (1 to 10).
- Gate: open time (seconds), denied message time.
- Data: snapshot retention (days), button to clear the log (with confirmation).

States: Saved toast, validation error, backend does not support settings yet (read-only view).

5.5 Login, About, 404 (SHOULD)

Purpose: Complete the product feel.

Contents

- **Login:** email, password, sign-in button, error message, no public data before login.
- **About:** block diagram, decision rule in plain words, list of technologies, team names.
- **404 / Offline:** short message and a button back to Live Gate.

States: Wrong password, session expired.

REUSABLE PARTS

6 Component Library

Component	Description	Variants / states
Top bar	Brand mark with hazard stripe, nav links, connection badge, clock, user label	Active link, mobile menu
Status hero	Camera frame and the big status word	Standby, checking, granted, denied, offline
KPI tile	Mono label with orange tick and a large number	Count-up on change
Hour chart	12 stacked bars with hour labels	Empty hour, hover value
Missing bars	Label, track, count	Zero state
Event row	Time, verdict, detail	New (highlight), granted, denied
Snapshot card	Image, time, chip	Loading, missing image, selected
Chip	Small bordered label with text	OK, bad, neutral
Buttons	Primary (orange fill), secondary (outline), segmented filter	Hover, focus, disabled
Table	Header row, zebra optional, row hover	Sortable, empty
Lightbox	Large photo with details and close button	Keyboard close (Esc)
Toast	Short confirmation at top right	Success, error
Empty state	Message plus next step	No data, no matches, offline
Connection badge	LIVE with pulsing dot, OFFLINE, DEMO DATA	3 states

7 Data, Realtime and Backend

Source data (do not change): one row per decision in `logs/events.csv`.

Column	Example	Meaning
timestamp	2026-11-02 14:32:05	Local time of the decision
result	VIOLATION	COMPLIANT or VIOLATION
helmet / vest	True / False	Whether each item was seen
missing	vest	helmet, vest, helmet;vest or none
snapshot	snapshots/20261102_143205.jpg	Photo path (violations only)

Endpoints

Endpoint	Returns	Priority
GET /api/events	Last 300 rows as JSON (already built in server.py)	MUST
GET /snapshots/<file>	Violation photo	MUST
GET /api/stats?range=24h	Totals and hourly counts computed on the server	SHOULD
GET /api/stream	MJPEG video with detection boxes (needs a small change to main.py)	SHOULD
GET and POST /api/settings	Read and write detection and gate settings	LATER
POST /api/login	Session cookie for the officer account	SHOULD

Realtime: the page polls `/api/events` every second. Rows are de-duplicated by timestamp + result + missing, and only new rows animate. A later upgrade is Server-Sent Events so the server pushes new rows.

Gate status rules (front end): latest event COMPLIANT and under 5 s old = ACCESS GRANTED. Latest event VIOLATION and under 7 s old = ACCESS DENIED. Otherwise STANDBY. These match `OPEN_SECONDS` in the Python code.

Animation specification

Trigger	Effect	Timing
Access granted	Barrier arm lifts with a slight overshoot; green lamp on; panel tints green	0.9 s, spring ease
Access denied	Red lamp pulses, panel shakes sideways once, tints red	0.45 s shake
KPI value changes	Number counts from old to new	0.7 s ease-out
New log row	Slides down from above with a yellow flash that fades	0.7 s
Chart update	Bars grow in place to new heights (elements are updated, not recreated)	0.7 s
Status change	Panel background cross-fades	0.4 s
Reduced motion	All animation and transitions disabled	Always respected

WORKS ON EVERY SCREEN

8 Responsive and Accessibility

Width	Layout
1200 px and up	Two columns: main (status, KPIs, chart) and side (log, missing)
900 to 1199 px	Two columns, tighter spacing
Below 900 px	One column in this order: status, KPIs, log, chart, missing, snapshots
Below 480 px	KPIs in a 2 by 2 grid, log rows stack, nav becomes a menu button

Accessibility checklist

- Text contrast 4.5:1 or better. Use the lighter red for red text.
- Status is always written, never colour alone. Use blue or orange alongside red and green where possible.
- The status word sits in an `aria-live="polite"` region so screen readers announce changes.
- Visible focus ring (2 px orange) on every link, button and input.
- Real buttons and links, labelled inputs, alt text on snapshots, semantic table markup.
- Respect `prefers-reduced-motion` and the dark or light system setting.

Security and privacy

- Require login for everything except the demo mode. Bind the server to 127.0.0.1 unless it is deliberately shared.
- Violation photos can contain faces. Keep them for a limited time (for example 7 days), restrict who can view them, and tell workers a camera is in use.
- No secrets or keys in the front-end code. Use HTTPS if it is ever hosted.
- The site only reads the log. It cannot open the gate, which keeps the safety logic in one place.

9 Tech Stack and File Structure

Recommended: plain HTML, CSS and JavaScript (no framework), served by the small Python `server.py`. It has no extra installs, is easy to explain in a viva, and runs on any laptop.

Path	Purpose
main.py	Detection and gate control (unchanged)
server.py	Serves the pages, /api/events, /snapshots and later /api/stream
web/index.html	Live Gate page
web/reports.html, web/events.html	Report and history pages
web/settings.html, web/login.html, web/about.html	Supporting pages
web/css/app.css	Design tokens and shared component styles
web/js/app.js	Polling, de-duplication, count-up, row animation, status logic
web/assets/	Logo mark, favicon, inline SVG icons, demo placeholders
logs/events.csv, snapshots/	Data written by main.py

Assets to prepare: logo mark (orange square with hazard stripe), favicon, three web fonts (Barlow, Barlow Condensed, IBM Plex Mono) with system fallbacks, a small inline SVG icon set (camera, check, cross, download, filter, clock), and a few sample snapshot images for demo mode.

Build order

Step	Task	Done when
1	Shared app.css with tokens, top bar, cards, chips	Top bar looks identical on two test pages
2	Live Gate page reading /api/events, with demo mode	Rows appear live; badge shows LIVE or DEMO DATA
3	Status logic, barrier animation, count-up numbers	Granted and denied states switch correctly
4	Violation Report page with snapshots and CSV export	Export downloads a valid file
5	Event History table with filters and pagination	Search finds a known row
6	Login, Settings, About, 404	Pages linked from the nav
7	Live camera stream (MJPEG) and polish	Camera panel shows boxes in real time
8	Responsive pass and accessibility check	Usable at 390 px wide; keyboard only works

EXACT WORDING AND FINAL CHECKS

10 Microcopy and Test Checklist

Situation	Text
Nobody at the gate	STANDBY. Waiting for next person
Checking	Checking PPE...
Granted	ACCESS GRANTED. Helmet and vest verified
Denied	ACCESS DENIED. Missing: vest
Backend unreachable	OFFLINE. Showing last known data
Demo mode	DEMO DATA. Start server.py to see live events
No violations	No violations recorded for this period.
Search found nothing	No checks match these filters. Clear filters

Final test checklist

- Each state (standby, checking, granted, denied, offline) displays correctly and with a text label.
- Run main.py: a real decision appears on the page within about a second.
- A violation photo shows in the report and in the lightbox.
- CSV export opens in Excel with the right columns.
- Stop server.py: badge changes to OFFLINE, then recovers when restarted.
- Works at 1440 px, 1024 px and 390 px wide, in light and dark mode.
- Everything is operable with the keyboard only.
- Demo rehearsed in the real room, with a backup screen recording.

Scope note: the current working build covers the Live Gate page and a basic report. The other pages in this blueprint are the full product vision. For the demo, build the MUST pages first and describe the rest as future scope.